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Procedia Computer Science 233 (2024) 558–569

Procedia Computer Science

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5th International Conference on Innovative Data Communication Technologies and Application (ICIDCA 2024)

Implementation of Efficient Machine Learning Techniques for Prediction of Cardiac Disease using SMOTE

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Abstract

The leading cause of mortality in the world for the past few decades has been cardiovascular disease (CVD) or cardiac arrest, which is caused by conditions that affect the Heart. With Machine Learning (ML), predictive models are refined and adapted to test health industries in a more efficient way. ML algorithms will helpful to make an early prediction for those who may need to make a change to their lifestyle and receive appropriate medical treatment. Our research presents supervised ML classifiers capable of detecting significant features using machine learning techniques to predict Heart disease. An over sampling method is presented here for learning from unbalanced datasets. Several classification techniques are employed with different features in our prediction model. Performance metrics are analyzed to determine their effectiveness when applied to the UCI heart dataset using ML techniques by SMOTE oversampling*;* A maximum of 96.6 percentage of accuracy, 90 percentage of sensitivity, and 100 percentage of specificity and precision were achieved by the Random Forest method.

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Peer-review under responsibility of the scientific committee of the 5th International Conference on Innovative Data Communication Technologies and Application

Keywords: Random Forest Classifier, Machine Learning Techniques, Disease Prediction, Feature Elimination, Cardiovascular Disease.

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10.1016/j.procs.2024.03.245

1. Introduction

Diagnosing Heart Disease (HD) can be difficult due to several factors that can increase the risk of acquiring a cardiac condition. Diabetic complications, high cholesterol levels, abnormal heart rates, high blood pressure, and other factors can be classified as modifiable or non-modifiable factors [1]. For determining the severity of HD, people have used a variety of data mining techniques. A heart attack is a very complex illness, so it is essential to treat it appropriately. To identify different metabolic syndromes, medical science, and data mining are used. Failure to do so would cause a heart attack or death. Those at risk of heart disease should seek the advice of cardiologists to determine the best treatment as early as possible to prevent the occurrence of any undesirable events. This paper explains how classification data mining for predicting HD plays an important role in cardiovascular prediction and data analysis [2]. A major challenge to modern health care is the provision of quality services and accurate diagnoses. In this work, we make efforts to diagnose these conditions in the heart in advance to avoid catastrophic consequences. Data from the medical field can be analyzed and extracted to provide information that is important for health care. An extensive medical data set can be analyzed using data mining techniques to uncover hidden information.

Machine Learning is part of data mining and can identify and predict various diseases with its large scale of well-designed data. In medicine, machine learning can help identify and predict various diseases. The application of machine learning techniques to the diagnosis of heart diseases has been investigated several studies. A major goal of this paper is to help doctors diagnose HD at the earliest possible stage. This will lead to effective treatment for patients and minimize side effects. A machine learning algorithm is very important for discovering and analyzing hidden patterns in data. After data analysis, ML techniques are used to make a diagnosis of HD more precise [3], [4]. This work analyzed the effectiveness of Random Forest (RF), Naive Bayes (NB), AdaBoost (AB), Decision Tree (DT), Support Vector Machine (SVM), K-Nearest Neighbor (KNN), and Logistic Regression (LR) for predicting early HD. An oversampling method called Synthetic Minority Oversampling Technique (SMOTE) is commonly used to resolve imbalance issues. Randomization balances the distribution of classes by replicating minority class examples. This is accomplished by synthesizing new minority instances from those already existing. A linear interpolation method is used for the minority class to generate the virtual training records [5]. helps improve the performance metrics of ML.

2. Related Work

In this section, we examine work being done by ML Techniques to predict heart disease. Mohan et al., [1] developed a model, HRFLM, which combines a random forest and linear regression model. This model provides an 88.7% accuracy level in HD prediction. Golande et al., [6] described a method with different ML algorithms for classifying HD. The authors made a comparative study on KNN, K-Means, and DT algorithms using classification accuracy. Comparing the Decision Tree with other ML algorithms, this work concludes that it has the highest accuracy value. In [7], As a result of using the complete set of attributes as a classifier, the author developed the sequential minimal optimization (SMO) classifier and obtained an accuracy of 86.468 percent. Alotaibi has examined various ML models, comparing different algorithms [8]. The Rapid Miner tool in this study offers effective results using the decision tree algorithm. According to [9], they built three classification models to predict HD, such as DT, LR, and Gaussian nave Bayes (GNB). The models used a strategy for reducing single-value decomposition functions that reduced the number of attributes from 13 to 4. The Prediction Model obtains 82.75% of the accuracy values for LR and GNB. This study developed a HD prediction model utilizing the clinical dataset from the UCI Repository consisting of 297 cases and 13 attributes using a user-friendly heart disease prediction system (HDPS) to examine heart disease prediction. The HDPS was developed as a new approach to classifying HD. It has been shown to be close to 80% accurate in classifying HD when using this approach. In [10], they propose a novel multimodal disease risk prediction algorithm based on convolutional neural networks (CNN-MDRP) which makes use of both structured and unstructured hospital data in order to produce an accurate prediction. According to the authors in [11], using the UCI dataset of 297 instances and 13 characteristics, Orange and Weka data mining tools were used to develop a system to predict HD based on the outcomes of the following criteria: Naive Bayes, RF, KNN, and SMO classifiers, which were evaluated based on their accuracy and recall. A

study published by Kumar et al. [12] trained five machine learning classifiers that were used to predict HD based on this dataset containing 303 instances and ten attributes. The classifiers used were SVM, LR, DT, KNN, and RF. Among the three classifiers, Random Forest was able to achieve an accuracy of 85.71 percent, which was more significant than the others. Taking into consideration the research described in earlier articles, it became evident that the primary goal of the proposed research would be to construct a cardiac prediction system utilizing the Table 1 data to make a prediction regarding the heart. In this study, we have used the Python package to classify the HD dataset using the various Machine Learning algorithms. It is compared regarding performance metrics, such as accuracy, precision, sensitivity, and specificity.

3. Materials And Methods

This section comprises Supervised ML classifiers, proposed research methodology, Dataset description, Data pre-processing, and Performance metrics.

3.1. Dataset

The term data set refers to an individual file that contains multiple records all of which have been gathered for a particular purpose. This research was conducted using the heart data set from the UCI Machine Learning Repository [13]. The heart data set contains two classes (0 and 1) where 0 reflects that the individual does not have heart disease and 1 represents that the individual does; 303 instances and 14 attributes (columns) are represented in this Table 1 along with the blood pressure, heart rate, cholesterol, depression level, Angina value and Chest pain of the individual. According to the Dataset, it can be assumed that women face a lower probability of heart disease than men. Heart diseases must be diagnosed accurately in order to treat them effectively. As a result, traditional approaches to prediction and diagnosis are not capable of providing accurate results.

Table 1. Attributes Description for the Heart Disease.

Features	Features Description	Type of Feature
Age	The patient's age in years	Numeric
Sex	The gender of the patient should be described	Nominal
Cp	As there are many forms of chest pain	Nominal
Trestbps	The Resting blood pressure	Numeric
Chol	Cholesterol level of the patient	Numeric
Fbs	Fasting blood sugar level in the patient	Nominal
Resting	The results of the electrocardiogram	Nominal
Thalach	The maximum heartbeat	Numeric
Exang	An Exercise induces angina	Nominal
Oldpeak	The patient's level of Depression	Numeric
Slope	Patient condition during peak exercise	Nominal
Ca	Fluoroscopy Results	Numeric
Thal	The status of Heart	Nominal
Num	Target value (Prediction Attribute)	Nominal

$$\text{If person} = \begin{cases} coronary\ heart\ disease & 1 \\ otherwise & 0 \end{cases} \quad (1)$$

3.2. Data Pre-Processing

There are 303 patient records in the dataset, and six records are found with missing values. Data pre-processing involves calculating the mean of particular columns and putting them in place of missing values for those six records. A pre-processed dataset contains 297 patient records and the missing values have been replaced on six records. As described in equation (1) above, if a person suffers from coronary heart disease, then the target value for the equation must be set to 1 while if not, it needs to be set to 0, indicating that the body does not suffer from coronary heart disease. After pre-processing 303 patient records, pre-processing indicates that 138 patients had heart disease, whereas 165 patients did not [14]. Table 1 provides a detailed description of features.

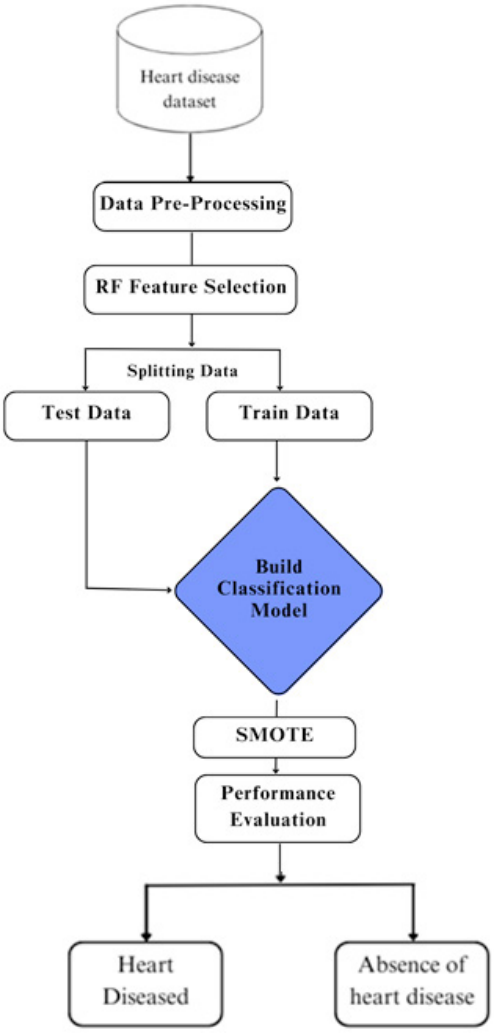


Fig. 1. Experimental diagram for identifying heart disease.

3.3. Machine Learning Models

This study made use of a variety of supervised machine learning algorithms to accomplish its goals. For the purpose of practicing the fundamental algorithm of supervised machine learning, the labeled training dataset is employed to train the algorithms. This qualified model is then loaded into a non-labeled testing research dataset to categorize it into related categories [15]. This section of the article will provide a brief overview of the proposed ML formulas to detect diseases based on supervised learning algorithms

3.3.1. Decision Tree

A DT is a very specific form like a flowchart. A decision tree is chosen because, compared to other ML techniques, they offer fast and reliable results; very little data preparation is necessary, and easy to interpret [16], [17]. Both classification and regression can be performed using Decision Trees using Classification and Regression Trees (CART algorithm). Find the paths to the leaf nodes from the root node decision tree represent classification rules in equation (2) used to calculate tree values.

$$EV = (FPO \times LO) + (SPO \times LO) - Cost \tag{2}$$

3.3.2. Logistic Regression

LR uses the logistical function of the outcome of probabilistic values between 0 and 1. In order to implement LR as a binary classification, a threshold value is determined, which defines the separation of the data into two groups [18], [19]. The data can be divided in LR using a straight line or a hyperplane by the use of equation (3).

$$g(E(y)) = \alpha + \beta x1 + \gamma x2 \tag{3}$$

3.3.3. Naive Bayes

The NB classifier utilizes the Gaussian distribution rule to perform the classification analysis in accordance with the classification concerns of the problem in hand [20], [21]. NB determined by the number of prior and posterior conditional probability classes shown in equation (4). Also, it is highly scalable, simple to use, and rapid to forecast.

$$P(A|B) = (P(B|A) P(A)) / P(B) \tag{4}$$

3.3.4. Adaboost

AB is widely used in an ensemble learning model which is based on ensemble learning. This is referred to as Adaptive Boosting. AB is able to integrate multiple weak classifiers into a solid classifier; it is intended to improve the learning ability of weak learners. A new weight is assigned to each instance, and weights assigned to incorrectly classified instances are higher.

3.3.5. Random Forest

The RF integrates several decision trees to obtain the best results. A bootstrap aggregation is used primarily for tree learning. Several random samples are created based on an entire data set**.** Row sampling is used to generate every model from the Bootstrap samples. During the training process, every model is independently trained. Based on the combination of all models, the final outcome is decided by a majority vote.

3.3.6. Support Vector Machine

An SVM uses mathematical models to deal with classification and regression problems, depending on their nature. Statistically based methods for finding the most optimal hyperplane are used in the SVM algorithm. With the help of the kernel trick, it transforms your data and then finds an optimal boundary based on these transformations.

3.3.7. K-Nearest Neighbors

KNN is a statistical learning algorithm that is old and simplest to use. The K value specifies the number of nearest neighbors to be used [22], [23]. The KNN determines how the new input should be classified by using the similarity between the new input and existing input samples in its training set using a Euclidean distance calculated by the equation (5). The KNN approach is the most basic algorithm for solving classification and regression issues.

$$d = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]} \quad (5)$$

3.4. Synthetic Minority Oversampling Technique

The SMOTE technique uses oversampling to overcome class imbalances in medical data. In SMOTE, random synthetic data is generated for the minority class based on the closest neighbors of the minority class. The SMOTE algorithm adopts KNN approach. Here K is the nearest neighbor, integrating the data and then making the synthetic samples in the space. It computes the distance between the feature vectors and their nearest neighbors. The difference in the two features is multiplied by a random number (0, 1), after which it is summed up to this attribution. One of the pioneer algorithms is SMOTE, from which many other algorithms are derived

3.5. Feature Importance

Machine learning relies heavily on feature importance for analysis. The coefficient value determines the importance and ranking of each feature. The experiment was done using clinical data collected from the UCI repository. It is mainly applied to biomedicine. The important features were identified using Random Forest feature ranking algorithm. From a dataset containing 13 features, this study identifies the features that are most relevant based on their feature importance.

3.6. Performance Metrics

An early detection model for Cardiovascular Disease based on the UCI dataset is built using seven classification techniques. The classification accuracy of several learning approaches is compared [24]. Furthermore, these learning approaches accommodate accuracy, precision, recall/sensitivity, and specificity [25]. Based on our confusion matrix, we calculate the performance metrics that are used to measure our performance. A confusion matrix includes elements such as True positive (TP): which indicates that the patient has a disease and that the test results are positive. False Positive (FP): when the patient does not have any disease, but testing shows there is a positive result even though there is no disease. True Negative (TN): when the result is negative and the patient is healthy. False negative (FN): it occurs when a test does not reveal the presence of a disease. When analyzing medical data, false negatives are crucial. The performance metrics are described below.

- Accuracy:
  $$\text{Accuracy} = TP+TN / TP+FP+FN+TN \quad (6)$$
- Precision:
  $$\text{Precision}=TP / TP+FP \quad (7)$$
- Recall/Sensitivity:
  $$\text{Sensitivity}=TP / TP+FN \quad (8)$$
- Specificity:
  $$\text{Specificity}=TN / TN+FP \quad (9)$$

4. Results and Discussions

The experimental design for the proposed approach is shown in Fig. 1. Phase one of the proposed work incorporates data collection from the UCI repository. Data pre-processing includes the detection of missing values and replacing them with mean values for better prediction accuracy. For early detection of heart attack, the generated data will be further studied and analyzed utilizing classification algorithms after data pre-processing as depicted in Algorithm.1.

Algorithm.1:
Step 1: Collect heart data from the UCI repository in .csv format.
Step 2: Perform Data pre-processing to handle missing values and remove duplicate values.
Step 3: Split the data into two sets: one for training and the other for testing.
Step 4: For the purpose of data analysis, the training data fed into the ML classifier algorithms.
Step 5: Identify the essential features by Random Forest Feature Ranking Algorithm.
Step 6: Introduce the SMOTE Technique for handling imbalanced Data.
Step 7: Analyze the result and calculate performance metrics.

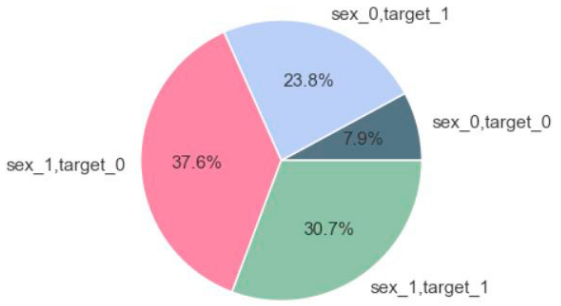


Fig. 2. Composition of Sex and Target value in the dataset.

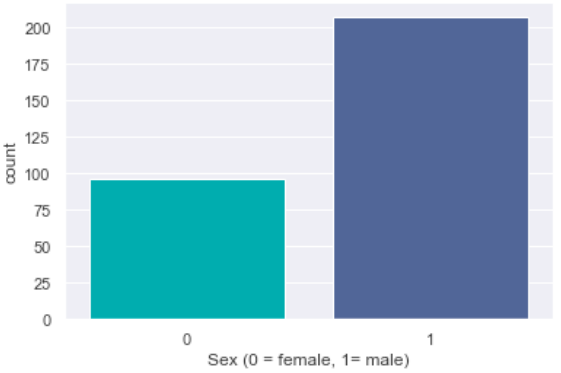


Fig. 3. Gender wise distribution of the Dataset.

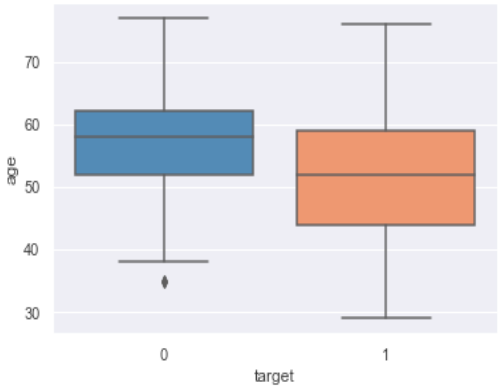


Fig. 4. Age distribution of patient with ± heart disease

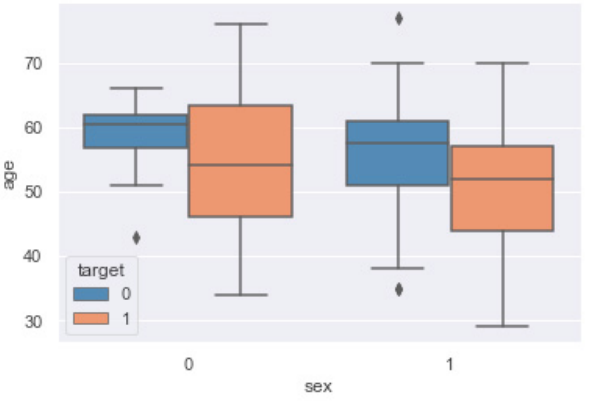


Fig. 5. Target Value over the sex

Fig. 2 illustrates the plot of the composition of sex and target value in the HD dataset. The graph plotted before the prediction result is approximately 200 male test cases and 100 female test cases, as shown in Fig. 3.The attributes in Table 1 represent an entry in different ML algorithms. The input data is purposely split into two groups: 80 percent is from the training data set, and 20 percent is from the test data set. During the model's training process, the training set will be used as the input data. The test dataset is used to determine the performance of a trained model based on

the results of the training. Performances were compared and analyzed in each algorithm based on the various parameters. Fig. 4 represents target values with age attributes. Classification based on age concludes that the age between 50 and 65 has no substantially reported problem of a heart attack. Thus, the target value is 0. Whereas this study observed that if the person is between 45 and 55 has a significantly higher risk of having a heart attack with a target value of 1. Thus, we obtain that the highest and lowest with respect to the targets are observed in the age range of ~45-65. Here, Fig. 5 shows target values over the sex attribute in our dataset, and this proves that females have a maximum risk of a heart attack in age ranges of ~45-65, whereas males in the age range of ~45-55 have the maximum probability of getting heart attack and the least prone are in the age range of ~50-62.

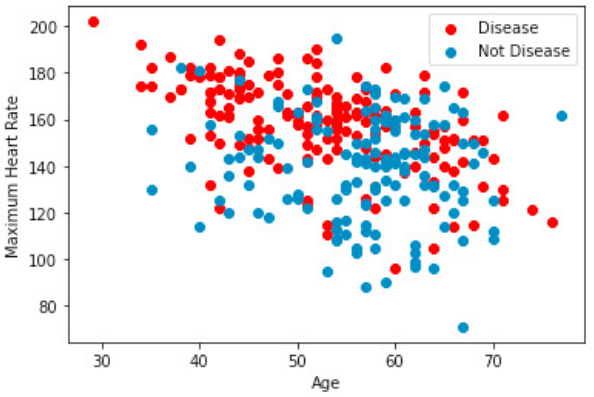


Fig. 6. Scatter plot for Thalach value.

In accordance with the scatter plot shown in figure , Graphical representation for maximum heart rate (Thalach) in this study, i.e., age vs Thalach value (as X-Y graph), the incidence of heart attack is represented by red plot, and the blue plot shows the absence of heart attack. According to the results, people who have a maximum heart rate are more likely to suffer from a heart attack than those with a regular heart rate. The age is greater than 55, and the maximum heartbeat is nearly 160; they are healthier than those who have a lower heart rate. Likewise, the result shows that if an individual is under 45 years of age and the rate of heartbeat is higher than 120, then the chances of getting a heart attack are extremely high.

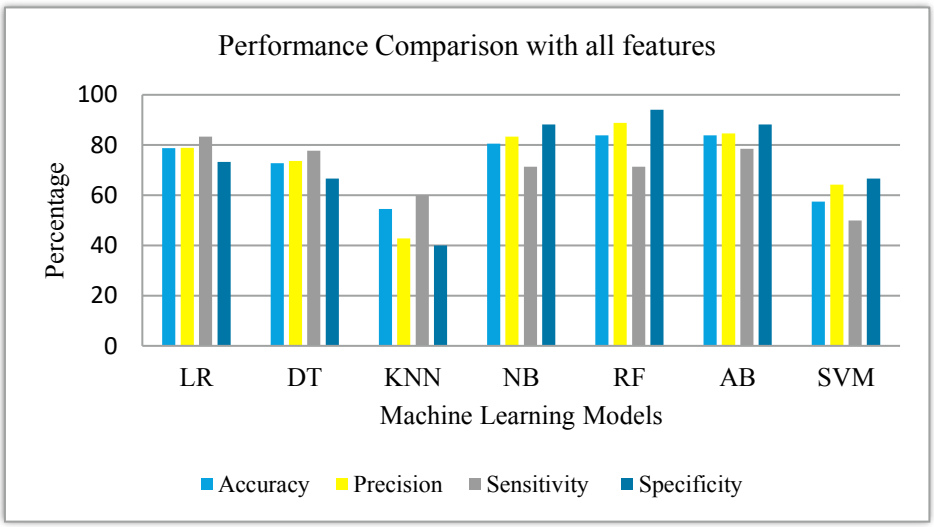


Fig. 7. Classification performance on all features

Table 2. Classification results of Machine Learning Techniques using all features without SMOTE.

ML	Accuracy	Precision	Sensitivity	Specificity
LR	78.7	78.9	83.3	73.3
DT	72.7	73.6	77.7	66.6
KNN	54.5	42.8	60.0	40.0
NB	80.6	83.3	71.4	88.2
RF	83.8	88.8	71.4	94.1
AB	83.8	84.6	78.5	88.2
SVM	57.5	64.2	50.0	66.6

Table 3. Classification results of Machine Learning Techniques using eight significant features without SMOTE.

ML	Accuracy	Precision	Sensitivity	Specificity
LR	83.8	84.6	78.5	88.2
DT	81.4	88.8	80.0	82.3
KNN	55.5	57.1	66.6	52.9
NB	81.8	80.0	78.5	73.3
RF	88.8	90.9	80.0	94.1
AB	87.0	91.6	78.5	94.1
SVM	67.7	75.0	67.1	88.2

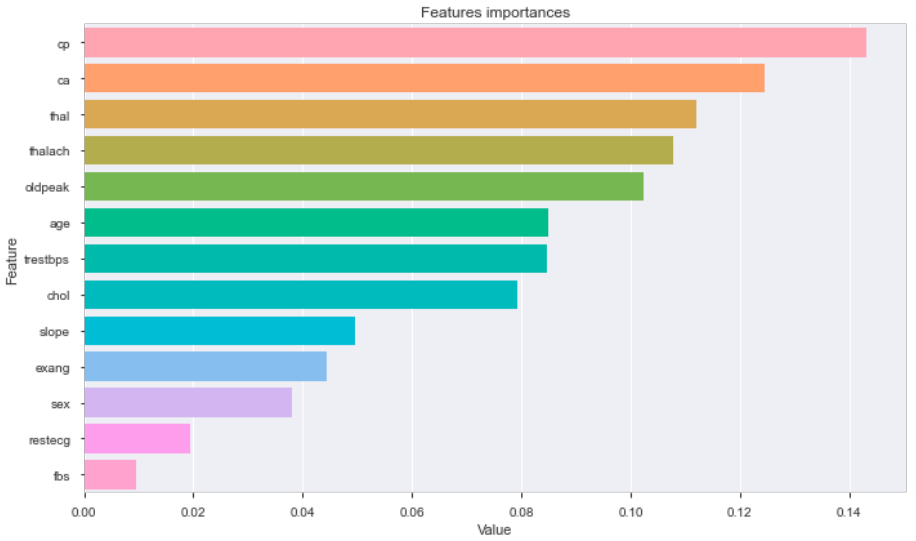


Fig. 8. Feature analysis by Random Forest Feature Ranking Algorithm.

Table 2 compares the performance of different classifiers using all 13 attributes from the heart dataset. The heart disease dataset is analyzed using ML algorithms, and a comparative study is made with the complete set of attributes. In Fig. 7, all models with all features are compared in terms of performance. A feature ranking algorithm was employed using Random Forest. The RF feature ranking algorithm predicts the importance of features based on Fig. 8. According to RF, the most relevant features include cp, ca, thal, thalach, oldpeak, age, trestbps, chol. Table 3 compares the performance of different classifiers using eight relevant attributes of heart disease without SMOTE.

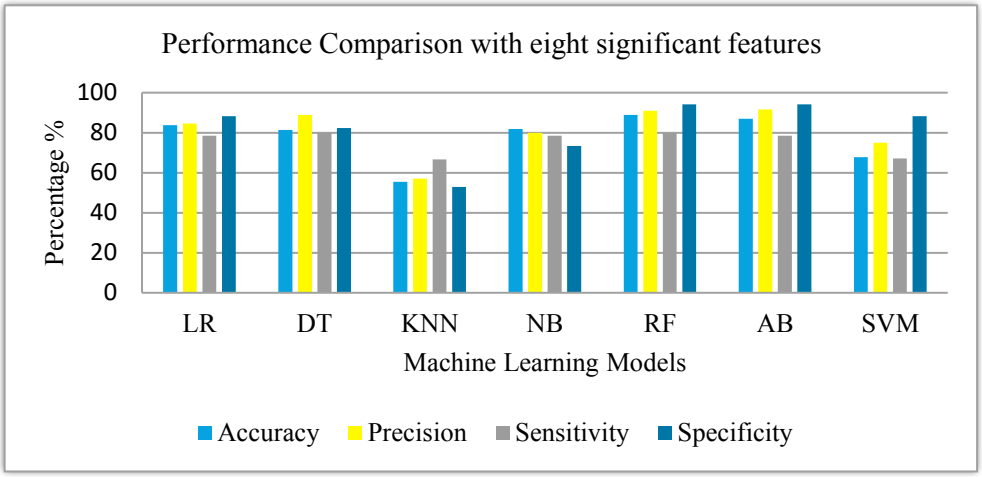


Fig. 9. Classification performance on eight significant features.

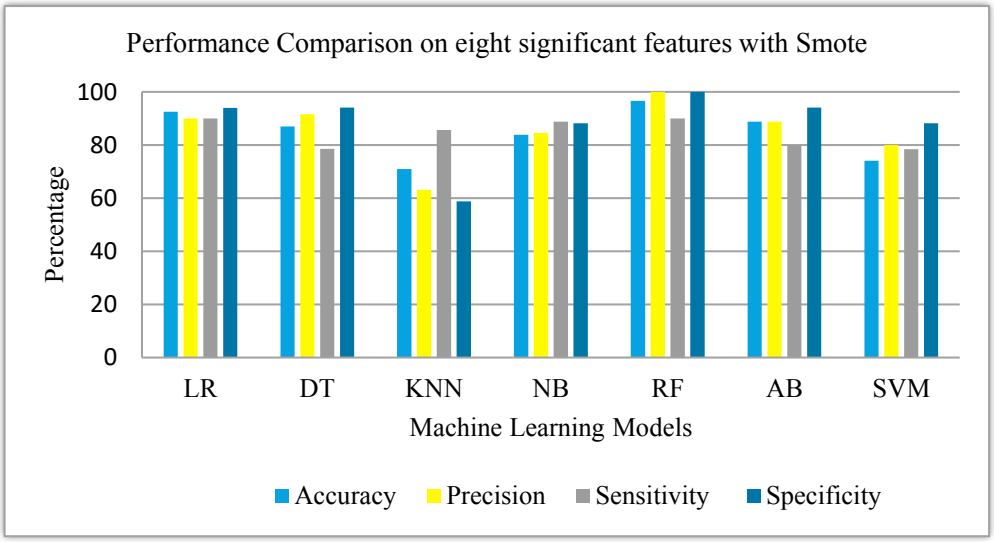


Fig. 10. Classification performance on eight significant features with SMOTE.

Table 4. Classification results of Machine Learning Techniques using eight significant features with SMOTE.

ML	Accuracy	Precision	Sensitivity	Specificity
LR	92.5	90.0	90.0	94.0
DT	87.0	91.6	78.5	94.1
KNN	70.9	63.1	85.7	58.8
NB	83.8	84.6	88.8	88.2
RF	96.6	100	90.0	100
AB	88.8	88.8	80.0	94.1
SVM	74.1	80.0	78.4	88.2

Fig. 9 describes the bar chart of classification performance and compares using machine learning techniques with

the top eight relevant features identified by the Random Forest feature selection method. Table 4 compares the performance of different classifiers using eight relevant attributes of heart disease with SMOTE oversampling technique. According to Fig 10, an analysis of seven different ML techniques for understanding the accuracy in relation to a heart attack for different age groups and sex. Amongst the seven techniques, we obtain the RF learning technique has able to achieve the maximum accuracy percent using eight significant features with SMOTE oversampling technique. Fig 11 shows the accuracy comparison of the various classifiers using all 13 features, eight important features with and without SMOTE. In terms of predicting patients' survival, Random Forest had good performance when used with SMOTE.

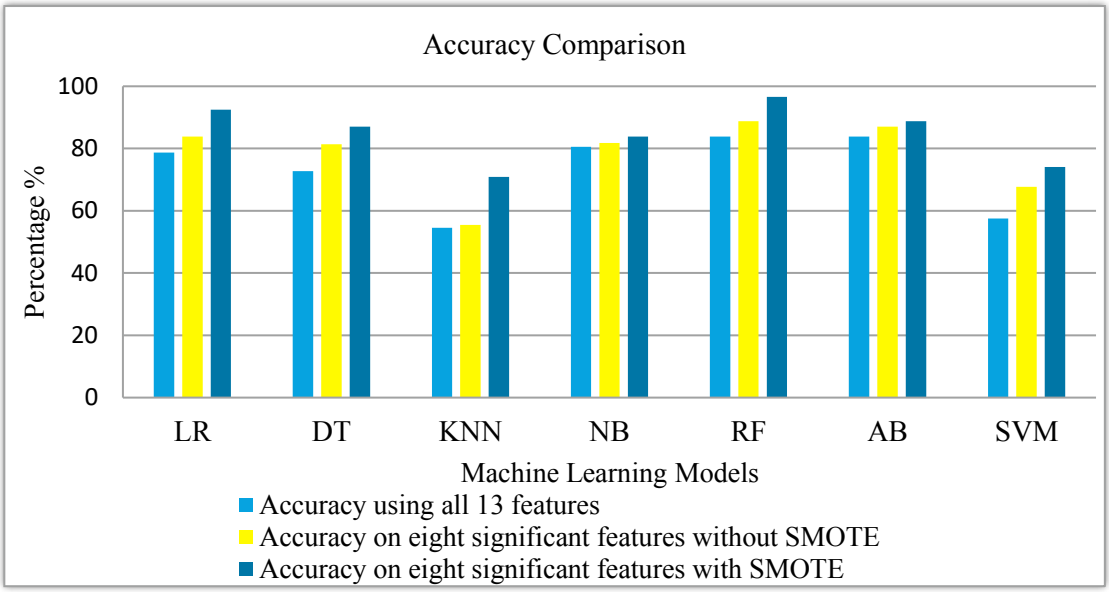


Fig. 11. Accuracy comparison of ML classifiers using eight significant features with and without SMOTE.

5. Conclusion

The work presented multiple supervised ML classification approaches that predict cardiac disease at an early stage using oversampling SMOTE as a pre-process for reducing data unbalance. The Random Forest feature selection method makes it possible to develop better strategies for selecting features from the heart disease dataset. Compared to other machine learning techniques, with Smote technique Random Forest achieves the most balanced outcomes in terms of precision and specificity of 100%, as well as 90% sensitivity. Based on the outcome prediction value, the accuracy is 96.6% which is an excellent result. In the future, we plan to Implement IoT sensor-based prediction system that will be more helpful for better prediction. Using IoT sensor resources, feature extraction will be made easier as well as data collection and measurement issues will be clarified with the help of IoT sensor.

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